

Sep 7 — 5 papers

Episode 1 · September 7, 2026 · 5 papers

Risk-Aware Optimal Control with Rulebooks

READ

Tichakorn Wongpiromsarn

It turns a priority-ordered rulebook of driving requirements into a lexicographic optimization over how much each rule's risk exceeds its allowed threshold, and solves it with an anytime branch-and-bound that hands you a policy plus a certified bound on how suboptimal it is.

PROBLEM Rulebooks give you a priority ordering over driving rules, but real evaluation of those rules is uncertain, and existing formulations do not say how to trade a small violation of a high-priority rule against uncertainty in whether it was violated at all. The open piece is anytime computation with a trustworthy certificate: safety-critical planners need a bound on suboptimality at whatever moment you stop the solver, not just an answer at convergence.

RESULTS The abstract reports no numbers at all. Two evaluations are named: a synthetic benchmark with a known optimum, which is the honest way to check whether the certified gaps are real, and a highway-merging simulation with CVaR-based collision, rear-braking, headway and comfort rules. No baselines, no runtimes, no gap magnitudes, no traffic densities are quoted, so the empirical strength of this work is unassessable from the abstract alone.

WHY IT MATTERS Read it for the formulation, not the application. Your maneuver coordination work has exactly this structure: safety dominates comfort, comfort dominates efficiency, and every input arrives with uncertainty attached, especially fused detections from collective perception where occlusion means you are reasoning about objects you have not directly observed. Excess risk above a per-rule threshold is a cleaner way to encode that than tuned scalar weights. The critical limitation for you is that this is single-agent optimal control. There is no negotiation, no shared intent, no message exchange, so it does not give you a coordination mechanism. But if you want a principled way to rank candidate joint trajectories once a coordination protocol has proposed them, or to reason about how perception uncertainty should propagate into a lane-change decision, the machinery transfers.

METHOD Each requirement gets a risk measure and an acceptable risk threshold, and each induces a risk-evaluation function mapping a policy to the risk of violating that requirement. Excess risk above threshold becomes the objective, and the priority relation makes it lexicographic, so lower-priority rules only choose among policies already optimal at every higher level. The algorithm is filtering plus branch-and-bound that progressively tightens a certified optimality gap and, importantly, characterizes the whole set of near-optimal policies at each priority level rather than one point, since that set is what the next priority level searches over.

CAVEAT The guarantee has a visible seam. Gap validity holds for any finite budget, but convergence to zero holds only under additional assumptions the abstract does not state, and those assumptions are where the real content lives. A sharp reviewer would press on whether they require finite policy classes or known distributions, and whether the risk-evaluation functions are assumed exactly computable. If CVaR is estimated from samples, the certificate is over estimated risk, not true risk. Everything is simulation, one merging scenario, no field data.

arXiv:2609.05199v1 PDF notable

Open-Loop Stackelberg LQ Difference Games with Coupled-Affine Inequality Constraints: An Exact OCP--LCS/LCQP Reformulation

SKIM

Shrestha Ghosh, Puduru Viswanadha Reddy

They show that a leader-follower linear-quadratic game with hard inequality constraints coupling both players can be rewritten exactly as a single optimal control problem with complementarity conditions built in, which finally makes the equilibrium computable.

PROBLEM Open-loop Stackelberg equilibria in unconstrained linear-quadratic games have clean solutions from coupled matrix recursions, but the moment you add inequality constraints that couple the states and controls of both players, the follower's best response turns piecewise and non-smooth and the leader is left with a bilevel problem that has no closed form. Earlier treatments smoothed, penalized, or restricted to equality constraints, so an exact characterization of the constrained case was still open.

RESULTS There are no numbers in the abstract at all. The evidence is a single illustrative constrained network-flow game, with no runtimes, no problem dimensions, no baseline method, and no accuracy comparison, which is a conspicuous gap for a letter whose whole selling point is enabling numerical computation.

METHOD They encode the follower's constrained optimality conditions as a discrete-time linear complementarity system, then cast the leader's problem as optimal control subject to that system. Eliminating the state trajectory and the dynamic multipliers collapses the whole thing into one large linear complementarity quadratic program over the leader's variables, with an explicit recovery map that reconstructs the follower's strategy afterwards. The point worth holding onto is that this is an exact equivalence under their assumptions, not an approximation, so what you compute is a genuine generalized Stackelberg equilibrium rather than the solution to a softened surrogate.

CAVEAT The program they reduce to is precisely the class that is computationally hard in general, and they describe it as large-scale themselves, yet nothing is said about which solver was used, how cost grows with the horizon, or whether the returned point is a global solution or merely a stationary one. The setting is also one leader and one follower, whereas intersection and merge coordination involve many followers whose constraints couple to each other, and the phrase under the stated assumptions is carrying real weight here, since exactness of this kind usually needs a unique follower response plus regularity conditions that a crowded constraint set will violate.

WHY IT MATTERS Structurally this is your merge and intersection problem wearing different clothes. A ramp vehicle deferring to mainline traffic, or an infrastructure manager fixing slots that vehicles then plan inside, is a leader-follower game, and collision avoidance and slot occupancy are exactly coupled affine constraints on joint states and controls, so the equilibrium concept and the exact reformulation are worth having in your toolkit. But there is no vehicle model, no communication channel, no delay or packet loss, no perception, and no real-time claim, so treat it as a formulation you cite for rigour rather than a method you can drop into a maneuver coordination stack. It has nothing to offer your collective perception thread.

arXiv:2609.04928v1 PDF incremental

Coupled Control and Wireless World Models for Resilient Remote Robotic Control

SKIM

H. P. Madushanka, Sumudu Samarakoon, Mehdi Bennis

They train a JEPA-style world model that predicts both a robot's future state and the wireless channel's future state from camera frames plus radio spectrograms, and use those joint predictions to decide when the robot can skip an uplink transmission.

PROBLEM A teleoperated robot streaming camera frames over a wireless link burns bandwidth and energy, and degrades when the channel drops out. Prior predictive schemes model robot dynamics or channel quality separately, so the scheduler never knows that a state prediction is about to become unreliable precisely because the link is about to fade.

RESULTS The abstract gives no numbers at all. It claims significant gains in communication efficiency, robustness and resilience while preserving navigation performance, against a PID controller, a model-free Deep Q-Network, and a Vision Transformer predictive baseline, all inside a synchronized Gazebo, ROS and Sionna co-simulation. Thirteen figures suggest the evidence exists in the body, but nothing is quantified up front.

METHOD Two coupled Joint Embedding Predictive Architecture world models learn in latent space, one over visual observations of robot dynamics and one over wireless conditions, the latter fed both raw spectrograms and a topological summary in the form of Persistence Images. Because both models forecast in the same embedding space, the controller can skip uplink transmissions when the predicted latent is trusted. They add a resilience mechanism that watches for divergence between predicted and observed latents and then adapts only the perception embedding, leaving the control policy untouched, which avoids full retraining when the channel or the visual scene shifts.

CAVEAT Everything lives in a ray-traced Sionna channel coupled to a Gazebo robot, so the wireless world model is being validated against the same propagation model that generated its training data, which is close to circular. The Persistence Image branch is the paper's most distinctive choice and the abstract offers no ablation showing it beats the raw spectrogram alone. The baselines are also mismatched, since PID and DQN are not communication-aware at all, so beating them on transmission count proves little.

WHY IT MATTERS Peripheral to your core three topics but genuinely relevant to your fourth. This is single-agent teleoperation, one robot and one link, so it says nothing about maneuver negotiation, collective perception fusion, or intersection arbitration. What it does offer is a concrete mechanism for the question underneath all of those, namely when a cooperative agent can safely stay silent because it can predict both its own next state and the channel's next state. If you are thinking about adaptive collective perception message rates or about how link degradation should feed back into a coordination controller, the coupled prediction idea and the Sionna plus ROS co-simulation setup are worth borrowing. Ignore the teleoperation framing.

arXiv:2609.04851v1 PDF notable

Ao Shen et al.

They train a robot to push through crowded corridors by keeping the pedestrians reactive during learning, and by letting each pedestrian be more or less willing to yield, which the robot has to figure out on the fly from how people are positioned relative to it.

PROBLEM Crowd navigation policies are usually trained against pedestrian trajectories that either ignore the robot entirely or assume every pedestrian yields the same amount, so the policy never learns the real uncertainty of not knowing who will give way. That gap stayed open because keeping robot-conditioned crowd reactions inside the training loop is expensive and because heterogeneous responsiveness has no clean label to feed the policy.

RESULTS The abstract gives no numbers at all. No success rates, no collision rates, no named baselines, no named crowd dataset, no density values. All we get is the claim of improved safety and robustness across response conditions and densities, transfer without retraining to different crowd-flow layouts, ablations supporting the gating and the two-stage encoding, and a real-robot run described only as operating with sparse observations.

METHOD The crowd simulator keeps pedestrian motion conditioned on the robot and assigns each pedestrian its own responsiveness level, and critically that responsiveness value is withheld from the policy input, so the network has to infer its consequences purely from robot-centered relative positions. The perception side is a two-stage gated Transformer that first encodes human-human relations and then human-robot relations, with a recurrent layer on top to carry interaction history across timesteps. Training uses a curriculum that steadily lowers pedestrian responsiveness so the crowd becomes progressively less cooperative.

CAVEAT The entire responsiveness story lives inside the authors' own simulator, so the headline result is that the policy handles the specific reactive pedestrian model they wrote. Real pedestrians are not drawn from that distribution, and the real-robot section as described is a feasibility demonstration rather than a test of the heterogeneous-responsiveness claim, because there is no way to control or measure how responsive the actual humans were. A sharp reviewer would also ask whether the two-stage gated Transformer beats a plain attention encoder given the same reactive training crowd, since the ablations compare architecture variants but the reactive simulator is the confounded ingredient.

WHY IT MATTERS This is peripheral to your work and you can safely leave it. The one idea worth carrying over is the framing of heterogeneous, unlabeled cooperativeness: your maneuver coordination and intersection work usually assumes a vehicle that either sends a valid intent message or does not, whereas this paper treats willingness to yield as a hidden per-agent variable that has to be inferred from motion alone. That maps directly onto mixed traffic with unequipped vehicles and onto trust and misbehaviour detection, where a partner may be broadcasting intent it will not actually honor. But the mechanism here is implicit inference from relative position with no communication at all, which is the opposite of your explicit intent sharing and trajectory agreement setting, so there is no method to lift.

arXiv:2609.05300v1

PDF

incremental

Chongwen Dong et al.

Three simulated robots each run their own language model to write a navigation policy once per round, a bandit to pick how to revise it, and a Double DQN to actually steer, and the full stack reaches the goal in every trial with a median of 42 ticks.

PROBLEM Language model reasoning is far too slow to run at control frequency, so the authors attack the mismatch between round-level policy reasoning and tick-level actuation, and they do it without a central planner that dictates team actions. The open piece is how to let a language model revise a policy while a reinforcement learner is still converging underneath it, without the revisions wrecking the learner.

RESULTS The complete configuration reached the goal in all ninety robot-round records, with a median completion time of 42 ticks and a ninetieth percentile of 73.2 ticks, and its median was 25 to 39.1 percent below the three ablated configurations. Four configurations, thirty rounds each, one fixed simulation. The authors themselves label this descriptive configuration-level evidence, which is unusually candid and also tells you how little was established.

WHY IT MATTERS Peripheral to your work, and worth being honest about that. The coordination here is three robots posting text summaries to each other between rounds, with no notion of a maneuver coordination message, no shared perception, no trajectory agreement, no bandwidth or latency model, and no communication failure. That said, one idea transfers: separating slow negotiated intent from fast local control, with the negotiated policy entering the fast controller as a conditioning input and a prior rather than as a command. That is structurally the problem you face when a maneuver coordination message arrives at one hertz while your trajectory tracker runs at fifty. This paper does not solve it under any realistic communication assumption, but the layering is a reference point if you are designing that interface.

METHOD Each robot owns a three-layer stack. A language model writes and revises a policy at round boundaries only, an Upper Confidence Bound bandit chooses which refinement mode to apply to that policy, and a policy-conditioned Double Deep Q-Network selects the actual per-tick action from navigation state, the active policy parameters, and the language model's action prior. Coordination between the three robots is nothing more than a shared round summary of policies, outcomes and learning feedback, and the three robots deliberately use different language model backends. The whole thing is a NetLogo and Python simulation.

CAVEAT Three robots in one fixed NetLogo scenario is not a benchmark, and the authors concede it. There is no external baseline at all, only their own ablations, so the 25 to 39 percent median gap says the full stack beats its own crippled versions and nothing more. Ninety records from thirty rounds are not ninety independent samples, no confidence intervals or significance tests are reported, and the central claim in the title, that refinement does not destabilize local learning, is never measured directly. No learning curves, no variance of the Double DQN across refinement events, no cost or latency accounting for the language model calls.

arXiv:2609.05133v1 PDF incremental